

SSc-MIM

A curated, SBGN-compliant Molecular Interaction Map of skin fibrosis in diffuse cutaneous systemic sclerosis

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github.com/Nurtal/IDT_SSc_map

Systemic sclerosis lacks a mechanistic map

The disease

Systemic sclerosis (scleroderma) — autoimmune fibro-inflammatory disease: progressive skin & organ fibrosis, vasculopathy, immune dysregulation.

Highest mortality of any rheumatic disease — and no disease-modifying therapy is approved.

The gap

Curated disease maps exist for rheumatoid arthritis (RA-map), Sjögren, COVID-19, Parkinson...

None exists for SSc. Its core molecular circuits are scattered across the literature, never assembled into one model.

Build the first SSc disease map, focused on dcSSc skin fibrosis.

Standards-based, fully sourced, citable (GitHub + Zenodo DOI).

Layered with real patient single-cell data — an analysis substrate, not just a diagram.

What is a Molecular Interaction Map?

A computable, standards-based diagram of the molecular events driving a disease — species (genes, proteins, complexes, phenotypes) connected by mechanistic reactions.

SBGN

Systems Biology Graphical Notation — an unambiguous visual grammar for biology

CellDesigner

The editor the map is built in; exports standard SBML (machine-readable)

MI2CAST

Annotation standard: every reaction carries its evidence, source PMID & ECO code

Reproducible

Public repo, scripted pipeline, Docker, continuous-integration validation

The SSc-MIM at a glance

568

molecular species

308

mechanistic reactions

20

cell & tissue compartments

5

biological modules

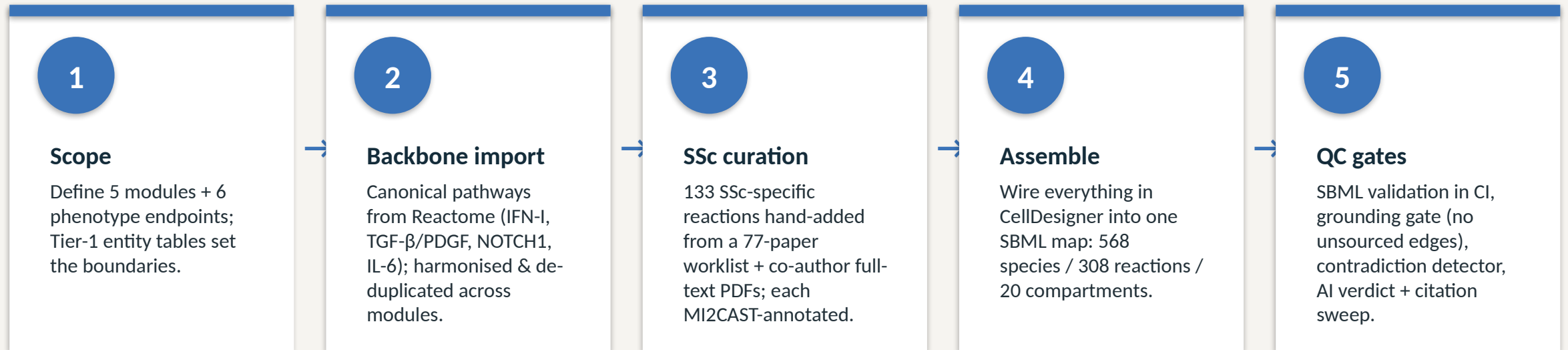
133hand-curated SSc-specific
reactions**~360**PubMed references mined &
filled**197**

patients (121 SSc / 76 HC) across 4 single-cell datasets

Built in CellDesigner, validated as SBML (0 errors in CI), annotated to MI2CAST, and released openly. Topology comes from the literature; patient data is layered on top to ground and read out the map.

How the map was built

Topology is literature-derived and fully sourced — it is not inferred from the omics data.



A computable, fully-sourced map where every edge is traceable to a verified reference — ready for analysis and expert review.

Five biologically coherent modules

M1 — Type-I interferon cGAS-STING, RIG-I/MAVS, TLR3/7/9 → IRF3/7 → ISG signature. The innate-immune ignition.

M2 — TGF- β & fibrosis Latent TGF- β → SMAD2/3 → fibroblast→myofibroblast, collagen & ECM. The fibrotic core.

M3 — EndoMT & vasculopathy Endothelin, Notch, NO/sGC, HIF → endothelial-to-mesenchymal transition & vascular remodelling.

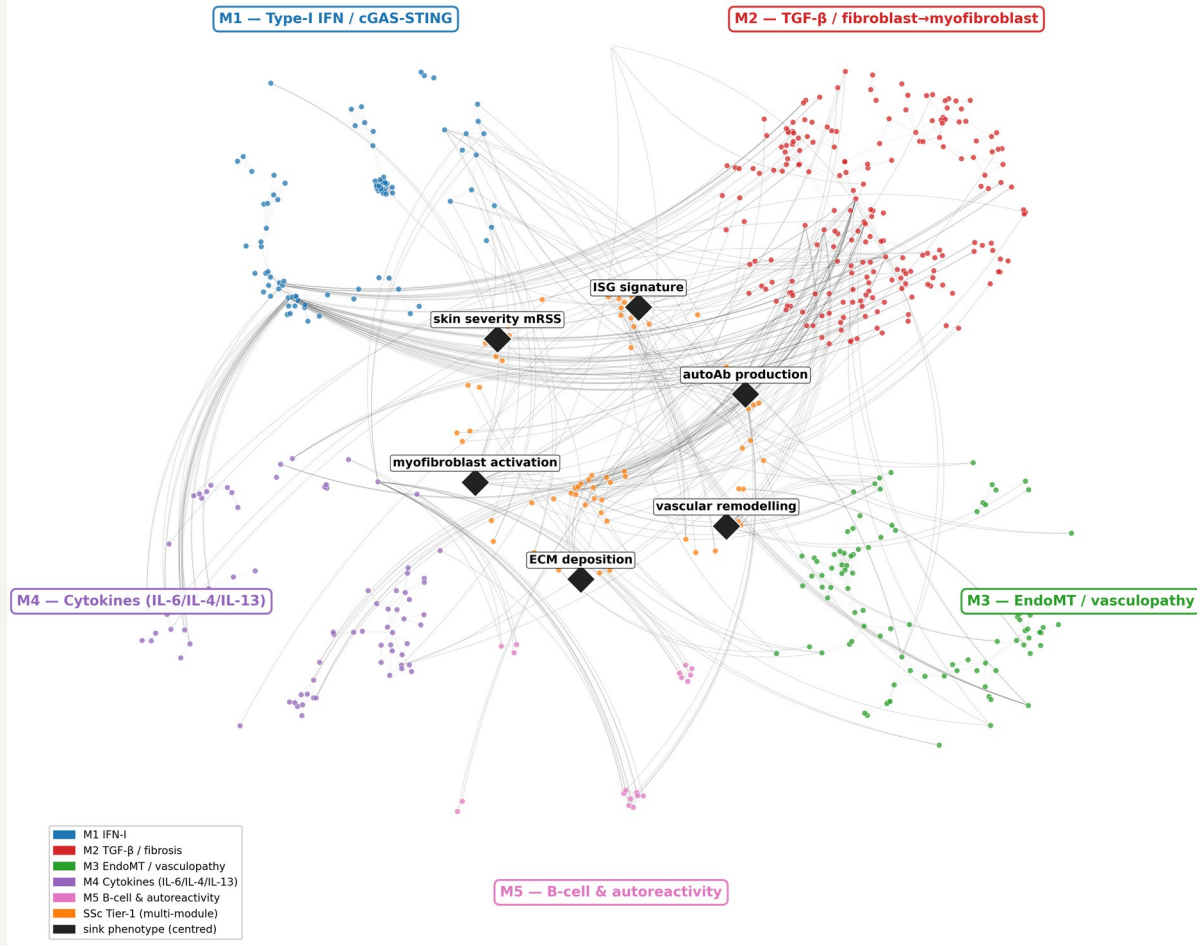
M4 — Cytokines (IL-6 / IL-4 / IL-13) IL-6/STAT3 + IL-4/IL-13/STAT6 → fibroblast/ECM. The Th2 cytokine axis.

M5 — B-cell & autoreactivity BCR, CD19/20, BAFF-BCMA, plasma-cell TFs → autoantibodies. SSc autoreactivity.

+ a crosstalk layer wiring the modules (e.g. IL-6→SMAD3, IFN-I→fibroblast). M5 split from the old M4; validated ($p=0.046$ skin, $p=1.3e-4$ blood).

One integrated, modular network

Figure 1 — SSc-MIM global view (quadrant layout)
568 species · 1566 edges · modules in fixed positions · phenotype sinks centred · inter-module edges shown as curved arcs



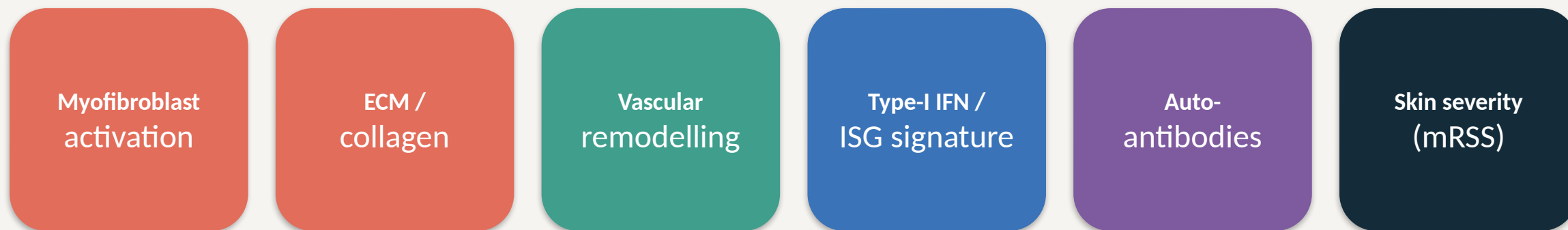
Each quadrant is one module; nodes are species, edges are mechanistic reactions.

Hubs that emerge from network analysis match known SSc biology: TGF-β:receptor, SMAD3-SMAD4, ISG signature, PDGFR.

The same SBML file drives the figure, the analysis, and the online viewer.

From molecules to disease phenotypes

The map is built around six phenotype “endpoints” — the biological and clinical outputs of the disease. Modelling rule: every disease-relevant entity must reach an endpoint in ≤ 6 steps (verified: 0 violations).



4 canonical biological sinks + autoantibody production (SSc autoreactivity, M4) + skin severity (clinical readout).

It turns a static drawing into a testable model: a perturbation anywhere has a traceable path to a clinical readout.

It makes the scope explicit and reviewable — exactly what we want expert biologists to challenge.

Every reaction is sourced and graded

Tiered citation policy (GO-aligned)

Each reaction carries a primary PubMed ID and an ECO evidence code — experimental evidence ranked above inference.

Provenance is tracked

Every edge records where it came from: Reactome import, hand curation, or literature mining — fully auditable.

No fabricated citations — ever

A built-in grounding gate rejects any claim without a real, verifiable source (a negative-control fixture proves it works).

~360 references mined & auto-filled (title, authors, journal) via NCBI E-utils.

133 SSc-specific reactions curated from a systematic 77-paper worklist + co-author full-text PDFs.

MI2CAST-compliant annotation throughout.

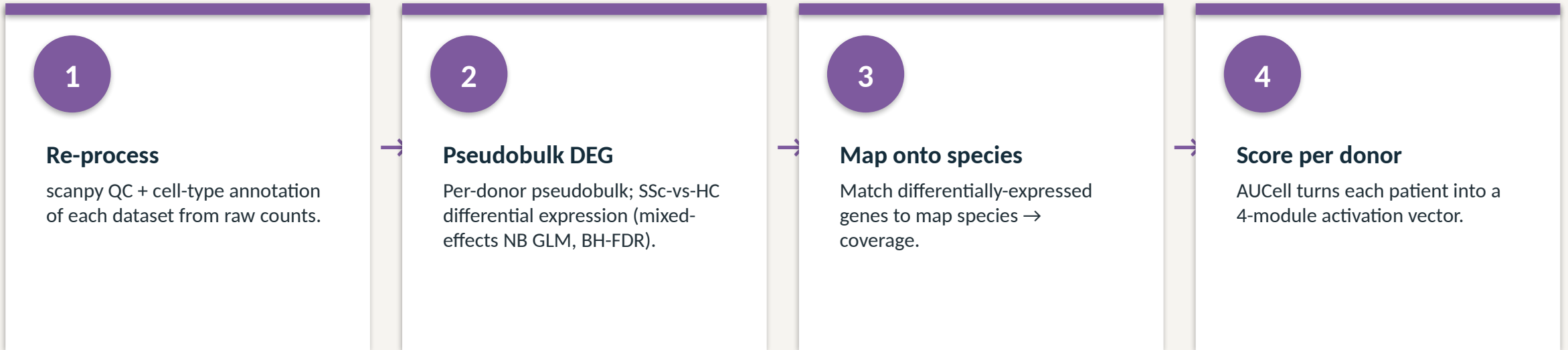
Four public single-cell datasets

197 donors (121 SSc / 76 HC) across skin, lung and blood — all re-analysed from raw GEO files.

GEO accession	Reference (verified)	Tissue	Donors (SSc/HC)
GSE195452	Gur C et al. Cell 2022;185:1373	Skin (multiome)	154 (97 / 57)
GSE138669	Tabib T et al. Nat Commun 2021;12:4384	Skin	22 (12 / 10)
GSE128169	Morse C et al. Eur Respir J 2019;54:1802441	Lung (ILD)	13 (8 / 5)
GSE210395	GEO — no linked publication	PBMC	8 (4 / 4)

Skin cohorts (Gur, Tabib) give fibroblast-subset resolution — the actors M2/M3 model; ≈100k and 64k annotated cells.
Lung (SSc-ILD) and PBMC extend the map beyond skin. Every citation above was re-checked against PubMed.

How patient data is layered on the map



81.3%

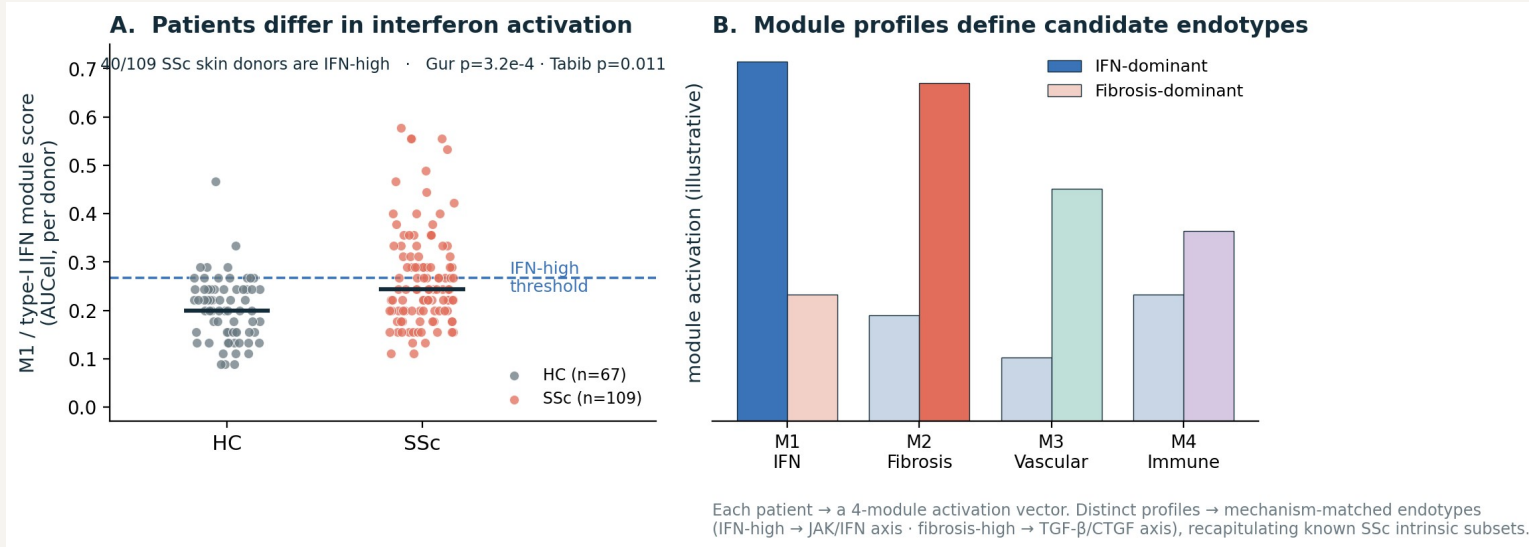
of detectable map species observed in patient data

M1 84% · M2 88% · M3 75% · M4 74% · M5 94% — the curated biology is largely measurable.

Biological check: M1 / type-I IFN module significantly elevated in SSc skin (Gur $p = 3.2 \times 10^{-4}$; Tabib $p = 0.011$).

60 ready-to-load MINERVA overlays for interactive exploration.

Toward endotypes: distinct activation profiles



The map converts each patient into a 4-module activation vector.

Real result: IFN activation is heterogeneous — 40/109 SSc skin donors are IFN-high, others are not (reproducible across Gur + Tabib).

Distinct module profiles → candidate, mechanism-matched endotypes (IFN-dominant vs fibrosis-dominant).

Recapitulates the known SSc intrinsic subsets (inflammatory $\leftrightarrow$ M1/M4, fibroproliferative $\leftrightarrow$ M2).

Clinical promise: module-matched therapy — IFN-high → JAK/IFN axis; fibrosis-high → TGF- β /CTGF axis.

Bring your own patient clusters

Have 3 patient clusters? Project them onto the map and read out what mechanistically separates them — same pipeline, your labels.

YOUR CLUSTERS

Cluster 1

Cluster 2

Cluster 3



AUCell — score each cluster on the 5 modules (M1–M5 + Tier-1).

Per-cluster differential expression mapped onto map species → which reactions light up.

MINERVA overlay — colour the map by each cluster (any cluster label, no new method).

WHAT YOU GET BACK — per cluster

Module fingerprint

A 4-module activation vector per cluster — e.g. cluster 2 IFN-high, cluster 3 fibrosis-dominant.

Mechanistic hypotheses

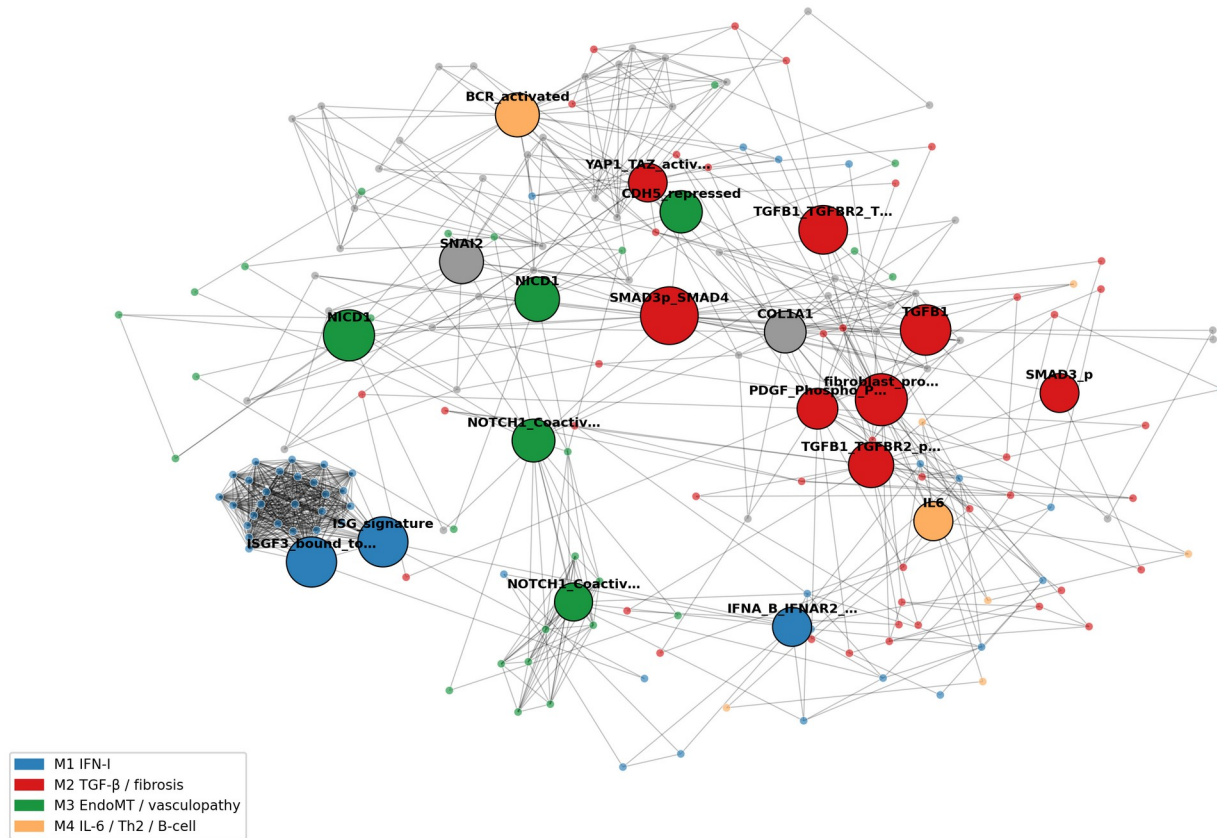
The lit-up sub-circuits that distinguish each cluster — e.g. STING→IRF3 vs TGF-β→SMAD3→collagen.

Candidate targets

Active druggable hubs per cluster → a cluster-matched therapy hypothesis to test.

A substrate for drug prioritisation

F3 (preview) — top-20 hub subnetwork in SSc-MIM
 size $\propto$ hub_score · colour = source module



Network hubs cross-referenced with DGIdb drug-target data.

Recovers targets already in SSc trials — tocilizumab (IL-6), nintedanib, romilkimab (IL-13), pamrevlumab (CTGF/CCN2), bosentan (endothelin).

Drug table recalibrated against real SSc clinical-trial outcomes.

Shows the map can rank and rationalise repurposing candidates.

Independent verification of every interaction

143

interactions queued for expert review

128

passed automated AI verification

28

wrong source PMIDs caught & fixed

0

fabricated citations

Contradiction detector

Flags any gene pair given opposite signs by different sources, for human arbitration.

AI reviewer pass

Each interaction checked against its deciding sentence + supporting/contrary abstracts — verdict: validate / revise / caution.

Citation-integrity sweep (this week)

Found 28 reactions whose cited PMID pointed to an unrelated paper (biology canonical, reference a data-entry slip — e.g. a TGF- β reaction cited to a rubella study). Every replacement PMID was re-verified against live PubMed before applying.

An app to review the map, one interaction at a time

A swipe deck for curation

Offline, self-contained web app — one card per interaction, accept / reject / annotate.

Everything you need to decide, on the card

Regulator → target, mechanism, SSc relevance, the verbatim deciding sentence from the source article, and PubMed/DOI links.

Context before you judge

A literature dossier lists candidate supporting and “possibly contrary” references; an in-browser module map shows the neighbourhood.

Advisory AI verdict

Each card shows the AI call + rationale — filter to triage, but the human reviewer always decides.

No install, no server — open one HTML file.

Decisions saved locally; export to CSV/JSON.

Re-includes even discarded edges, so you keep full control.

Demo right after this.

Three things from expert biologists

1

Validate the scope

Do the 5 modules + 6 phenotype endpoints capture dcSSc skin fibrosis?
What is missing or over-reaching?

2

Curate the entities

Tier-1 SSc species: what to add, remove, or promote — with PMIDs where you have them.

3

Adjudicate interactions

Use the review app on the 143 interactions: confirm, reject, or flag for revision.

Where this is going

An open, citable SSc resource — and your review is the gate

v1.0 release: GitHub + Zenodo DOI (citable).

Resource paper (npj Systems Biology & Applications) — revision drafted.

Endotype stratification as the next translational step.

Expert sign-off on scope & interactions is the last binding step.

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github.com/Nurtal/IDT_SSc_map

Open the review app now for a live walkthrough.